

NER Landslide Early Warning System - Parameter & Dataset Matrix

Complete Breakdown of Real vs. Synthetic vs. Mock Sources for SIH Panel Evaluation

System Readiness Overview: 45% Live Native APIs | 30% Calibrated Synthetic Data | 25% Simulated Schema Connectors

#	Data Stream	Current Source Type	Current Implementation File	Target Real-World Source	Status
1	GIS Basemap Tiles	REAL	Esri World Dark Gray REST (HeatmapViewer.tsx)	Esri / Survey of India NHP	DONE (Live)
2	Landslide Susceptibility (LSI)	SYNTHETIC (800)	XGBoost Model (baseline_xgb_model.joblib)	GSI Bhukosh NLSM 1:50k Vectors	DONE (Trained)
3	Landslide Inventory	SYNTHETIC	Sample Generator (gsi_inventory_loader.py)	GSI Bhukosh / NRSC Landslide Atlas	DONE (Schema)
4	Rainfall Telemetry (Nowcast)	SIMULATED SCHEMA	IMD Connector (ingestion/imd_fetcher.py)	IMD AWS Pune REST API (Mausam)	PENDING API KEY
5	24-48h Weather Forecast	ALGORITHMIC	Geotechnical Index (forecast_model.py)	IMD GFS 0.125 / NCMRWF Unified Model	DONE (Forward)
6	2-6h Surge Nowcast	NEURAL ARCH	2-Layer PyTorch LSTM (lstm_nowcast.py)	IMD Doppler Radar + In-situ sensors	AWAITING DATA
7	Digital Elevation Model	ALGO + SYNTH	Horn DEM Engine (dem_processor.py)	Copernicus DEM GLO-30 / CartoDEM	PROCESSOR READY
8	InSAR Ground Creep	SIMULATED SCHEMA	GEE Connector (gee_sentinel_fetcher.py)	Sentinel-1 SAR C-band IW SLC (ESA)	PENDING GEE KEY
9	NDVI / Moisture Indices	SIMULATED SCHEMA	Multispectral Pipeline (gee_sentinel_fetcher.py)	Sentinel-2 Optical 10m Bands	PENDING GEE KEY
10	Soil Moisture Saturation	CALIBRATED SIM	Soil Moisture Loader (soil_moisture_loader.py)	NASA SMAP L3 / In-situ TDR Probes	SIMULATED
11	In-Situ Geotechnical IoT	SIMULATED HOOKS	Threshold Hooks (lstm_nowcast.py)	SDMA / IIT-R Ground Slope Sensor Mesh	MOCK (Hardware)
12	Highway Corridor Graph	REAL GRAPH	NetworkX Graph (road_network_graph.py)	MoRTH / BRO Project Beacon & Swastik	DONE (Live)
13	Citizen Photo Reports	REAL	HTML5 Camera + Geolocation (ReportHazardModal)	Citizen Smartphone Crowdsourcing	DONE (Live)
14	Citizen Video Reports	REAL	HTML5 Media Capture API (ReportHazardModal)	Citizen Smartphone Edge Crowdsourcing	DONE (Live)
15	Offline Queue Protocol	REAL	IndexedDB + W3C Service Worker (offlineDb.ts)	Local Client Device Storage	DONE (Live)
16	Officer Ground Feedback	REAL	Feedback Modal + Retrain Pipeline (retrain_pipeline)	DDMA / PWD Field Engineers	DONE (Live)
17	Evacuation Audio Siren	REAL	Web Audio API AudioContext Synthesizer	Local Device Speaker Hardware	DONE (Live)
18	Spoken Voice Alert (TTS)	REAL	W3C Web Speech API (AlertBanner.tsx)	Browser Localized Speech Synthesis	DONE (Live)

Key Jury Defense Statement:

All ingestion modules in ml-engine/ingestion/ already conform to exact official API response schemas (IMD AWS, Copernicus Sentinel, GSI Bhukosh). Production activation requires only inserting API credentials into the .env file without changing any model inference logic.